

Video Game Programming: Game Design Fundamentals

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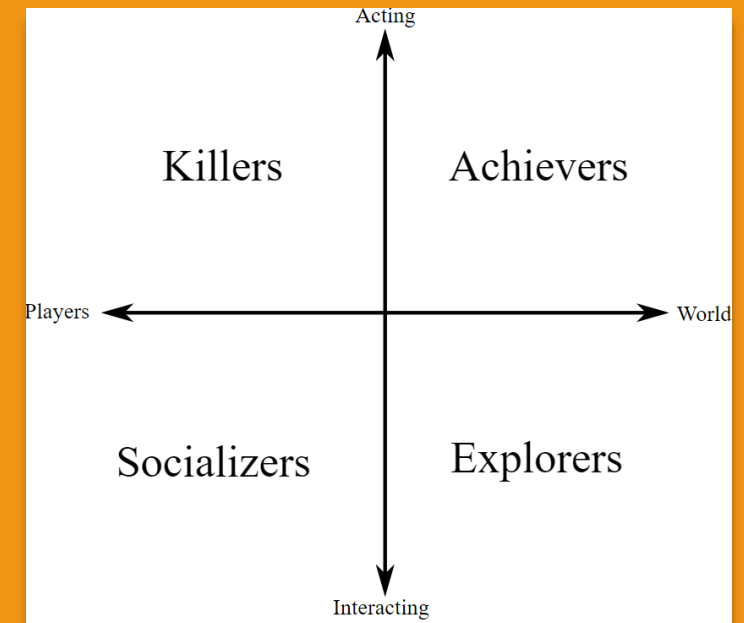
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What makes a video game (un)enjoyable?

- Fair / unfair
- Interactive / Can't do anything
- Clear goal / confusing
- Balanced / Unbalanced
- Right amount of player agency (i.e., player's ability to influence the game through their decisions and actions) / no or too much of it
- Nice and usable UI / ugly or unusable UI
- Cool graphics / poor graphics
- Catchy storing / no or boring story
- ...

Know your players

- Bartle taxonomy of player types (Bartle, “Hearts, Clubs, Diamonds, Spades: Players Who suit MUDs”, 1996)
 - Achievers prefer to gain points, collect rewards, and succeed in measurements
 - Explorers prefer discovering areas, immerse in the game world, at their own pace
 - Socializers play games for the social aspect, rather than the actual game itself
 - Killers thrive on competition with other players and prefer fighting them
- Proposed for multi-user dungeons (MUDs), but used also for single-player games
- Has several limitations, but still a starting point (Williams, Yee, Caplan, “Who plays, how much, and why? Debunking the stereotypical gamer profile”, J. Comp.-Mediated Comm., 13:4, 993-1018, 2008)



Video game genres

- 43 genres based on interactivity (Wolf, “The Medium of the Video Game”, 2001)
- Genres based on multiple variables (Aarseth, Sunnanå, Smedstad, “A multi-dimensional typology of games”, Digital Games Research Conference, 2003)
- Game website with more than 30 overlapping genres
- Linked to gameplay, not story or theme
- 4 main genres based on criteria for success (Egenfeldt-Nielsen, Smith, Tosca, “Understanding Video Games”, 2024)

	<i>Action games</i>	<i>Adventure games</i>	<i>Strategy games</i>	<i>Process-oriented games</i>
Typical action	Battle	Mystery solving	Build nation in competition with others	(Varies)
Criterion of success	Fast reflexes	Logic ability	Analyzing interdependent variables	Ambiguous

Video game platforms

- PC
 - Console (Xbox, PlayStation, Wii, Switch) and peripherals (e.g., Kinect)
 - Mobile
 - Web (e.g., WebGL)
 - VR headsets and tracked controllers
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- Cross platform: same game runs on different platforms, sometimes allowing for cross playing with players using other platforms
 - Running different parts of the game on different platform (e.g., FFXIII on PS1 and PocketStation, PlayLink games on PS4 and mobile)

Game elements

- List from Unity curricular framework:
 - Narrative: a string of connected events making up a story
 - Game design: how designer see the game
 - Objectives: players' goal or goals
 - Player options: what players can do to affect the game
 - Sequences: linking moments together for deeper, more interesting play
 - Mechanics: rules that control how players act in the game
 - Aesthetics: look-and-feel of a game
 - Dramatic elements: drawn from traditional novels, cinema, theater, etc.
 - Multimedia inclusion: text, still images, moving pictures, 2D, 3D, music, sound

The MDA model

- Hunicke, LeBlanc, Zubek, “MDA: A formal approach to game design and game research”, 2004
- Seeks to divide games into three separate dimensions:
 - **Mechanics:** rules and basic code, not what we see while playing, but what goes into constructing the game world, e.g., algorithms that determine reactions of NPCs
 - **Dynamics:** the way the game plays based on the mechanics; events that actually occur, or can occur, during the game, e.g., the behavior shown by NPCs; they are functions of mechanics, but may be surprising, as complex processes interact in (sometimes unpredictable) ways
 - **Aesthetics** covers the emotional responses evoked during interaction: Sensation (game as sense-pleasure), Fantasy (game as make-believe), Narrative (game as drama), Challenge (game as obstacle course), Fellowship (game as social framework), Discovery (game as uncharted territory), Expression (game as self-discovery), Submission (game as pastime)

Video game aesthetics

- All aspects experienced by the player, directly, such as audio and graphics, or indirectly, such as rules (Egenfeldt-Nielsen, Smith, Tosca, “Understanding video games”, 2024)
- Rules
- Geography and representation
- Number of players
- Narrative

Rules

- Can be very simple (e.g., in tic-tac-toe) or extremely complex (e.g., in RPGs)
- A game is its rules, but the game considered as a set of rules is not the experience of the game (Salen, Zimmerman, “Rules of Play”, 2004)
- Several ways to distinguish between different types of rules

Types of rules (Juul)

- Juul identifies three types of rules (Juul, “Half-real: Video games between real rules and fictional worlds”, PhD thesis, IT University of Copenhagen, 2003):
 - **Game state** rules cover the basic aspects of the game state (i.e., the exact condition at any point in time of all game elements)
 - **Outcome valorization** rules define which outcomes are considered positive and which outcomes are considered negative
 - **Information** rules determine what information the player receives during play about the game state

Types of rules (Salen and Zimmerman)

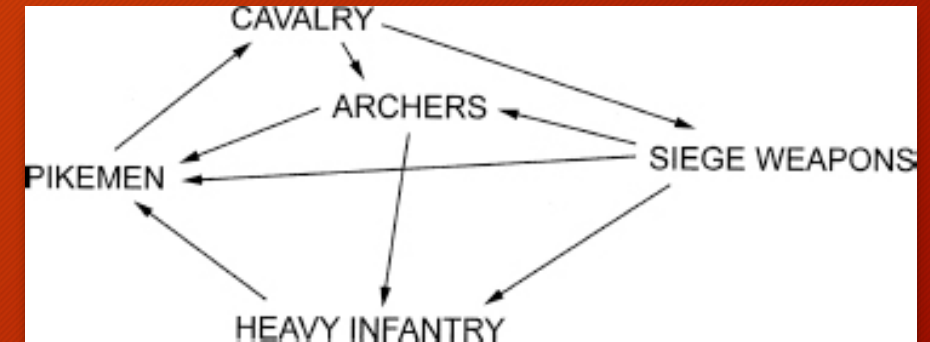
- Salen and Zimmerman have three types of rules (Salen, Zimmerman, “Rules of Play”, 2004):
 - **Operational** rules are what one would typically describe as the rules of a game; a combination of Juul’s “game state” and “outcome valorization” rules; they govern both a game’s processes and its conditions for victory
 - **Constitutive** rules are the underlying formal structures of a game that define its basic dynamics, e.g., logical and mathematical
 - **Implicit** rules are all the unwritten rules that we take for granted when playing a game

Types of rules (Egenfeldt-Nielsen, Smith, Tosca)

- Egenfeldt-Nielsen, Smith, Tosca propose two types of rules (“Understanding Video Games”, 2024):
 - **Interplay** rules determine the relationships and the properties of elements in a game; correspond to the physical laws of the gamespace; determine what can be done and, combined with player input, what happens
 - **Evaluation** rules decide which occurrences are rewarded and which are punished

Game balance

- Winning should be a function of player skill plus an element of randomness or luck, but should be unrelated to the game's initial conditions
- One classic way of achieving it is using **intransitive relationships** as in rock-paper-scissors game and in the fighting video game Destrega (1999) where Bidu (Fast) beats Est (Power), Est beats Foh (Span), Foh beats Bidu attacks, or directed graphs with cycles (in fig.)
- In strategy games with various civilizations with different strengths and weaknesses, the developers must plan carefully and test meticulously to ensure that all potential interactions are balanced and remain fair even across the different types of maps, regardless of players' strategies



Game balance in Age of Empires II (1999)

Geography and representation

- Geography is about the world where the video game is set
- It can add atmosphere, sense of realism, and sometimes influence players' possibilities (e.g., a blocking wall)
- Figure shows different geography and representation choices

2.5 isometric

Perspective	First person		Third person	
Dimensions	Two		Three	
Space type	Torus	Abstract	Free	
Off-screen Space	Dynamic	Static	None	
Scroll	Vertical	Horizontal	Free	None
Exploration	Forced	Free	None	

unconnected levels
zone-based multiscreen spaces
seamless multiscreen spaces

Graphical styles

- How the game express itself visually
- Graphical styles may be chosen to support the mechanics (or vice versa)
- Different choices:
 - Photorealism, using real actor filmed sequences or physically based rendering as in The Last of Us Part 2 or Horizon Zero Dawn
 - Caricaturism / cartoon style, presenting the essence of a person or object by exaggerating the most prominent features as in Crash Bandicoot, Zelda, and Dragon Quest series
 - Abstractionism, not representing real people or objects

Game audio

- **Vocalization** refers to the voices of characters, including voice-over speech and other off-screen elements
- **Sound effects** are sounds made by in-game objects (e.g., gun fire)
- **Ambient effects** are nonspecific sounds contributing to the game atmosphere (e.g., birds, distant sounds)
- **Music** is the soundtrack to the game, used to add atmosphere, set the mood, or attach a tune to an NPC
- Using recording or MIDI files
- Audio can now be affected by environment, physics, spatiality
- Audio can be part of the mechanics as in Loom, Guitar Hero, Rock band

Time

- From fast-paced arcade games to longer, slow playing sessions of some PC and console games
- Distinction between **play time**, i.e., the “real” time a player spends playing a game, and **event time**, i.e., the time that passes in the game world during this game (Juul, “Introduction to game time”, in “First person: New media as story, performance, and game”, MIT Press, 2004)
- Play time and event time can be 1:1, or there can be a fixed multiplier, or the relation can be altered in special moments (sped up during travel, cut-scenes, save & loading time)

Number of players

- **Single-player**

- Importance of artificial intelligence (AI)
- No need for Internet connection (unless to unlock additional content)

- **Multiplayer**

- AI is less important (unless in cooperative multiplayer games)
- Means to split screen or handle screen scrolling are need for local use
- Communication infrastructure is fundamental for remote use

Gameplay

- Game dynamics emerging from the interplay between rules and game geography (Egenfeldt-Nielsen, Smith, Tosca, “Understanding video games”, 2024)
- Crucial aspects for gameplay
 - Thinking games as interconnected systems
 - Player-centered design
 - Balance
 - Keeping challenge - Flow (Csikszentmihalyi, “Flow: The Psychology of Optimal Experience”, Harper & Row, 1990)
 - Level design to surprise as levels increase

Game design ToDo list

- Define game concept: description, platform, genre
- Script story and design geography & representation: world and characters, emotion, mood and theme
- Identify clear goals and objectives for players' progress, reward and motivation
- Decide if players will need reasoning, skills or both
- Invent or apply game mechanics to obtain dynamics: rules, balance, flow, gameplay
- Apply physics and mathematical models
- Design art and sound to connect to players and communicate game ideas
- Handle resource management because computing resources are finite

Bibliografy

- Egenfeldt-Nielsen, Smith, Tosca, Understanding Video Games: The Essential Introduction, 5th ed., Routledge, 2024
- Schreiber & Romero, Game Balance, CRC Press, 2022
- Gomila Salas Juan Gabriel, Borromeo Nicolas Alejandro, Hands-On Unity Game Development, 4th edition, Packt Publishing, 2024

